An introduction to computer modeling in earth system science.

EAS 6131. Ocean Modeling. 3 Credit Hours.

Developing and using ocean models from the simple shallow water to the full primitive equation model. Includes hands-on programming.

EAS 6132. Introduction to Climate Change. 3 Credit Hours.

The climate of the Earth, its radiation budget, greenhouse gases and their sources and sinks, potential changes due to anthropogenic activities, detection of climate changes.

EAS 6133. Marine Ecosystem Modeling. 3 Credit Hours.

Modeling population dynamics in the context of ocean circulation. Numerical techniques and simulation development.

EAS 6134. Inverse Methods and Time Series Analysis in Earth and Atmospheric Sciences. 3 Credit Hours.

Theory of data acquisition, time series analysis, and discrete inverse theory, with applications in the earth and atmospheric sciences.

EAS 6135. Introduction to Complex Environmental Systems. 3 Credit Hours.

Introduction to the concepts of environmental complexity through the inter-relationships between natural, human, and built systems.

EAS 6136. Paleoclimatology and Paleoceanography. 3 Credit Hours.

This course will explore the history of the Earth's climate, covering methods for reconstructing past climate and the mechanisms behind these climate changes.

EAS 6140. Thermodynamics of Atmospheres and Oceans. 3 Credit Hours.

The instabilities and flows created by exchanges of heat.

EAS 6142. Atmospheric Radiative Tranfser. 3 Credit Hours.